

MTH 166

Lecture-31

Directional Derivatives

Topic:

Vector Differential Calculus

Learning Outcomes:

1. To calculate directional derivatives
2. To find tangent plane to a surface

Directional derivative of a scalar field

Let $f(x, y, z)$ be a given scalar surface.

Let the given direction be $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

Then directional derivative of f in the direction of vector $\vec{b}$ is given by:

$$\begin{aligned} D_{\vec{b}}(f) &= \vec{\nabla} f \cdot \hat{b} = \vec{\nabla} f \cdot \left(\frac{\vec{b}}{|\vec{b}|} \right) \\ &= (f_x\hat{i} + f_y\hat{j} + f_z\hat{k}) \cdot \left(\frac{b_1\hat{i} + b_2\hat{j} + b_3\hat{k}}{\sqrt{(b_1)^2 + (b_2)^2 + (b_3)^2}} \right) \\ &= \frac{f_x b_1 + f_y b_2 + f_z b_3}{\sqrt{(b_1)^2 + (b_2)^2 + (b_3)^2}} \end{aligned}$$

Note:

1. Maximum rate of increase (Minimum rate of decrease) = $|\vec{\nabla} f|$

It occurs in its own direction.

2. Minimum rate of increase (Maximum rate of decrease) = $-|\vec{\nabla} f|$

It occurs in its opposite direction.

Problem 1. Find the directional derivative of the scalar function $(x^2y - y^2z - xyz)$ at the point $(1, -1, 0)$ in the direction $(\hat{i} - \hat{j} + 2\hat{k})$.

Solution. Let $f = (x^2y - y^2z - xyz)$

$$\Rightarrow f_x = (2xy - yz), f_y = (x^2 - 2yz - xz), f_z = (-y^2 - xy)$$

$$\vec{\nabla} f = (f_x \hat{i} + f_y \hat{j} + f_z \hat{k}) = (2xy - yz) \hat{i} + (x^2 - 2yz - xz) \hat{j} + (-y^2 - xy) \hat{k}$$

$$\text{At } (1, -1, 0): \vec{\nabla} f = -2\hat{i} + \hat{j} + 0\hat{k}$$

$$\text{Let } \vec{b} = (\hat{i} - \hat{j} + 2\hat{k})$$

$$D_{\vec{b}}(f) = \vec{\nabla} f \cdot \hat{b} = \vec{\nabla} f \cdot \left(\frac{\vec{b}}{|\vec{b}|} \right)$$

$$\Rightarrow D_{\vec{b}}(f) = (-2\hat{i} + \hat{j} + 0\hat{k}) \cdot \frac{(\hat{i} - \hat{j} + 2\hat{k})}{\sqrt{1+1+4}} = \frac{-2-1+0}{\sqrt{6}} = \frac{-3}{\sqrt{6}} \quad \textbf{Answer.}$$

Problem 2. Find the directional derivative of the scalar function (xyz) at the point $(1,4,3)$ in the direction of line from $(1,2,3)$ to $(1, -1, -3)$.

Solution. Let $f = (xyz)$

$$\Rightarrow f_x = (yz), f_y = (xz), f_z = (xy)$$

$$\vec{\nabla} f = (f_x \hat{i} + f_y \hat{j} + f_z \hat{k}) = (yz)\hat{i} + (xz)\hat{j} + (xy)\hat{k}$$

$$\text{At } (1,4,3): \vec{\nabla} f = 12\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\text{Let } \vec{b} = ((1 - 1)\hat{i} + (-1 - 2)\hat{j} + (-3 - 3)\hat{k}) = 0\hat{i} - 3\hat{j} - 6\hat{k}$$

$$D_{\vec{b}}(f) = \vec{\nabla} f \cdot \hat{b} = \vec{\nabla} f \cdot \left(\frac{\vec{b}}{|\vec{b}|} \right)$$

$$\Rightarrow D_{\vec{b}}(f) = (12\hat{i} + 3\hat{j} + 4\hat{k}) \cdot \frac{(0\hat{i} - 3\hat{j} - 6\hat{k})}{\sqrt{0+9+36}} = \frac{0-9-24}{\sqrt{45}} = \frac{-33}{3\sqrt{5}} = -\frac{11}{\sqrt{5}} \quad \textbf{Answer.}$$

Polling quiz

The del operator, denoted as $\vec{\nabla}$, is defined as:

(a) $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$

(b) $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} - \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$

(c) $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} - \hat{k} \frac{\partial}{\partial z}$

(d) $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} - \hat{j} \frac{\partial}{\partial y} - \hat{k} \frac{\partial}{\partial z}$

Problem 3. Find a direction that gives the direction of maximum rate of increase of scalar function $(3x^2 + y^2 + 2z^2)$ at $(0,1,2)$. Find the maximum rate too.

Solution. Let $f = (3x^2 + y^2 + 2z^2)$

$$\Rightarrow f_x = 6x, \quad f_y = 2y, \quad f_z = 4z$$

$$\vec{\nabla} f = (f_x \hat{i} + f_y \hat{j} + f_z \hat{k}) = 6x \hat{i} + 2y \hat{j} + 4z \hat{k}$$

$$\text{At } (0,1,2): \vec{\nabla} f = 0 \hat{i} + 2 \hat{j} + 8 \hat{k}$$

which is the direction of maximum rate of increase.

$$\text{Also, Maximum rate} = |\vec{\nabla} f| = \sqrt{0 + 4 + 64} = \sqrt{68} = 2\sqrt{17} \quad \textbf{Answer.}$$

Problem 4. Find a direction that gives the direction of minimum rate of increase of scalar function $(x^3 - xy^2 + y^3)$ at $(-2,1)$. Find the minimum rate too.

Solution. Let $f = (x^3 - xy^2 + y^3)$

$$\Rightarrow f_x = 3x^2 - y^2, \quad f_y = 3y^2 - 2xy$$

$$\vec{\nabla} f = (f_x \hat{i} + f_y \hat{j}) = (3x^2 - y^2)\hat{i} + (3y^2 - 2xy)\hat{j}$$

$$\text{At } (-2,1): \vec{\nabla} f = 11\hat{i} + 7\hat{j}$$

$$\text{Direction of minimum rate of increase} = -\vec{\nabla} f = -(11\hat{i} + 7\hat{j})$$

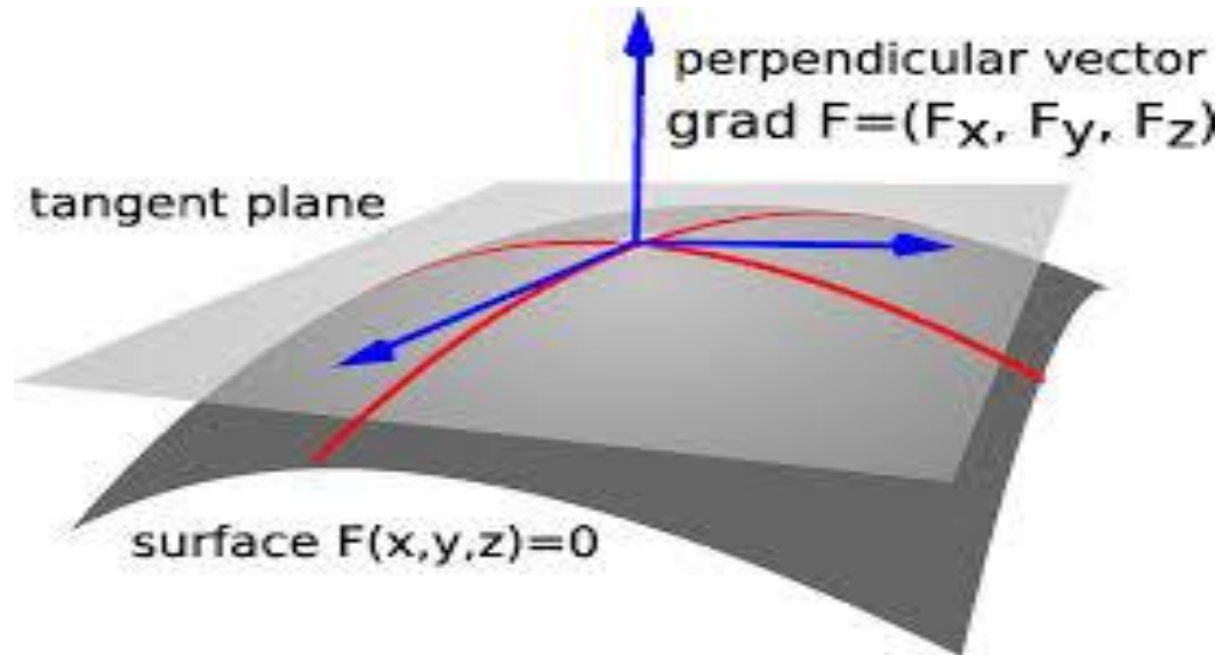
$$\text{Also, Minimum rate} = -|\vec{\nabla} f| = -\sqrt{121 + 49} = -\sqrt{170} \quad \textbf{Answer.}$$

Tangent Plane to a Scalar Surface

Let $f(x, y, z)$ be a given scalar surface.

Then equation of tangent plane to the surface $f(x, y, z)$ at the point $P_0(x_0, y_0, z_0)$ is:

$$(x - x_0)f_x(P_0) + (y - y_0)f_y(P_0) + (z - z_0)f_z(P_0) = 0$$



Problem. Find equation of tangent plane to surface $(x^2 - 3y^2 - z^2 = 2)$ at $(3,1,2)$.

Solution. Let $f = x^2 - 3y^2 - z^2 = 2$

$$\Rightarrow f_x = 2x, \quad f_y = -6y, \quad f_z = -2z$$

$$\text{At } (3,1,2): f_x = 6, \quad f_y = -6, \quad f_z = -4$$

Equation of tangent plane is given by:

$$(x - x_0)f_x(P_0) + (y - y_0)f_y(P_0) + (z - z_0)f_z(P_0) = 0$$

$$\Rightarrow (x - 3)(6) + (y - 1)(-6) + (z - 2)(-4) = 0$$

$$\Rightarrow 6x - 18 - 6y + 6 - 4z + 8 = 0$$

$$\Rightarrow 6x - 6y - 4z = 4$$

$$\Rightarrow 3x - 3y - 2z = 2 \quad \textbf{Answer.}$$

Polling quiz

The gradient of the scalar field function $f(x, y) = y^2 - 4xy$ at point (1,2) is:

- (a) $8\hat{i}$ (b) $-8\hat{i}$ (c) $8\hat{j}$ (d) $-8\hat{j}$

Polling quiz

The directional derivative of a scalar point function f in the direction $\hat{b}$ is given by:

(a) $f \cdot \hat{b}$

(b) $\text{curl}(f) \cdot \hat{b}$

(c) $\text{divergence}(f) \cdot \hat{b}$

(d) $\text{gradient}(f) \cdot \hat{b}$

